TOPS Tech Assignment-9

1. Explain the process of installing and configuring Hyper-V virtualization in Windows Server 2016.

Answer:

Hyper-V is a virtualization platform that allows you to create and manage virtual machines.

Installation Steps:

1. Open Server Manager.
2. Click Add Roles and Features.
3. Select Role-based installation.
4. Choose the server.
5. Select Hyper-V role.
6. Click Next and install.

Configuration Steps:

1. Open Hyper-V Manager.
2. Configure Virtual Switch (External, Internal, Private).
3. Create a New Virtual Machine:
    - Assign memory
    - Configure networking
    - Attach virtual hard disk
    - Install OS
4. Start and manage VM.

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2. How do you monitor server performance and manage event logs in Windows Server?

   Answer:

   Performance Monitoring:

   - Use Performance Monitor (perfmon).
   - Track CPU, memory, disk, and network usage.
   - Use Task Manager for quick monitoring.

   Event Logs Management:

   - Open Event Viewer.
   - Logs include:
     - Application
     - Security
     - System
   - Filter logs and analyze errors/warnings.
   - Create custom views for monitoring.

3. Describe the different types of storage options available in Windows Server.

Answer:

Types of storage options:

1. Direct Attached Storage (DAS) – Local disks.
2. Network Attached Storage (NAS) – File-level storage over network.
3. Storage Area Network (SAN) – High-speed block-level storage.
4. Storage Spaces – Combines multiple disks into pools.
5. Cloud Storage – Integration with cloud services.
6. iSCSI Storage – Block storage over IP network.

4.What is the role of File Server in Windows Server, and how do you configure it?

Answer:

Role:
A File Server stores and manages files and allows users to access them over a network.

Configuration Steps:

1. Open Server Manager.
2. Add File and Storage Services role.
3. Create a folder.
4. Right-click → Properties → Sharing.
5. Enable Advanced Sharing.
6. Set permissions (Read/Write).
7. Configure NTFS permissions for security.

5.Explain the process of implementing and managing Distributed File System (DFS) in Windows Server 2016.

Answer:

DFS allows sharing folders across multiple servers.

Steps:

1. Install DFS Namespace and DFS Replication roles.
2. Open DFS Management.
3. Create a New Namespace.
4. Add shared folders.
5. Configure Replication Group.
6. Set replication schedule and topology.

Management:

- Monitor replication.
- Add/remove servers.
- Check health status.

6. Discuss the built-in backup and recovery options available in Windows Server 2016 or 2019.

Answer:

Built-in options:

1. Windows Server Backup (WSB) – Full server and custom backups.
2. System State Backup – Includes registry, AD, etc.
3. Bare Metal Recovery – Restore entire system.
4. Volume Shadow Copy (VSS) – Snapshot backups.
5. File History (limited use).

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7.How do you configure Windows Server Backup to back up critical data?

Answer:

Steps:

1. Install Windows Server Backup feature.
2. Open Windows Server Backup.
3. Click Backup Schedule or Backup Once.
4. Select:
    - Full server or custom backup
5. Choose destination:
    - Local disk / external drive / network share
6. Set schedule (daily/weekly).
7. Finish setup.

8. Explain the steps for restoring files and folders using Windows Server Backup.

Answer:

Steps:

1. Open Windows Server Backup.
2. Click Recover.
3. Select backup location.
4. Choose date/time of backup.
5. Select Files and Folders.
6. Choose files to restore.
7. Select restore location:
    - Original or alternate
8. Start recovery.

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9. What are some common troubleshooting techniques for Windows Server startup issues?

Answer:

Techniques:

1. Use Safe Mode.
2. Check Boot Configuration (BCD).
3. Use Startup Repair.
4. Check hardware issues.
5. Analyze Event Viewer logs.
6. Use System Restore.
7. Repair using installation media.
8. Check disk using chkdsk.

10. How do you troubleshoot network connectivity problems in Windows Server?

Answer:

Steps:

1. Check physical connections.
2. Use commands:
    - ping
    - ipconfig
    - tracert
3. Verify IP configuration.
4. Check DNS settings.
5. Restart network adapter.
6. Disable firewall temporarily.
7. Check network drivers.
8. Verify routing and gateway.

11. Discuss common Active Directory-related issues and their troubleshooting steps.

Answer:

Common issues:

1. Login failures
    - Check user credentials and account lockout
2. Replication issues
    - Use repadmin tool
3. DNS problems
    - Verify DNS configuration
4. Group Policy not applying
    - Use gpupdate and gpresult
5. Domain controller not responding
    - Check services and connectivity

12. Explain how to troubleshoot performance problems on Windows Server 2016 or 2019.

Answer:

Steps:

1. Use Task Manager to identify resource usage.
2. Use Performance Monitor for detailed metrics.
3. Check CPU, RAM, Disk, Network usage.
4. Analyze Event Viewer logs.
5. Disable unnecessary services.
6. Update drivers and patches.
7. Check for malware.
8. Optimize applications and workloads.